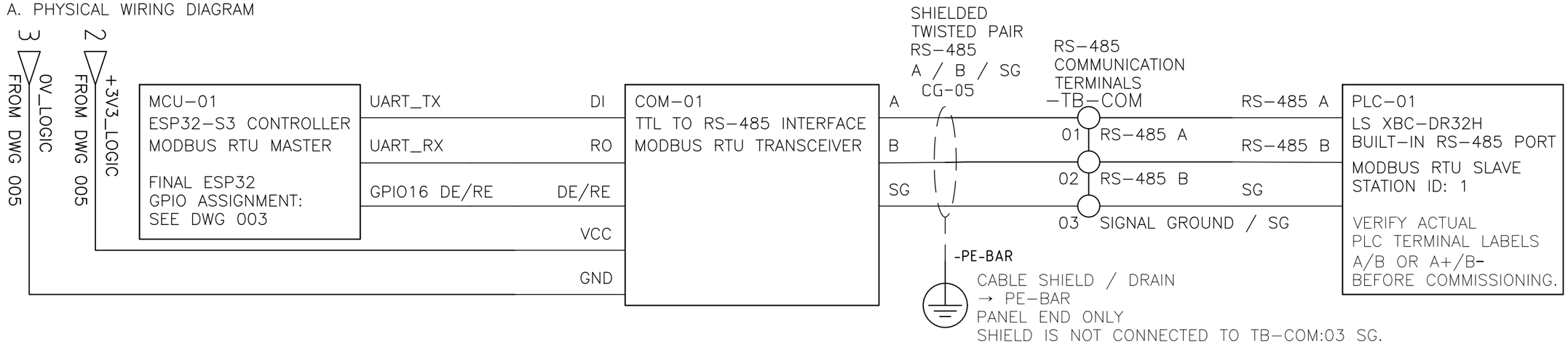
PLC-01 / MCU-01 MODBUS RTU RS-485 COMMUNICATION CIRCUIT

CONCEPTUAL PHYSICAL WIRING – VERIFY ACTUAL PLC AND RS-485 MODULE TERMINAL LABELS

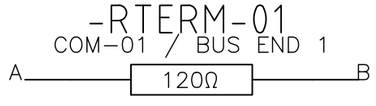
A. PHYSICAL WIRING DIAGRAM



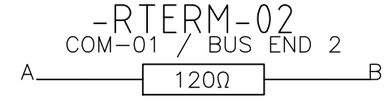
B. MODBUS RTU COMMUNICATION PARAMETERS

MODBUS RTU COMMUNICATION PARAMETERS		
Parameter	MCU-01	PLC-01
Protocol	Modbus RTU Master	Modbus RTU Slave
Station ID	Master	1
Physical Layer	TTL UART → RS-485	Built-in RS-485
Baud Rate	9600 bps	9600 bps
Data Bit	8	8
Parity	None	None
Stop Bit	1	1
Timeout	3000 ms	3000 ms
RS-485 Mode	Half Duplex	Half Duplex
Cable	Shielded Twisted Pair	Shielded Twisted Pair
NOTE: ALL COMMUNICATION PARAMETERS SHALL MATCH BETWEEN MCU-01 AND PLC-01.		

C. RS-485 TERMINATION / SHIELD DETAIL



VERIFY INTERNAL OR EXTERNAL
120Ω TERMINATION METHOD
BEFORE COMMISSIONING.



VERIFY INTERNAL OR EXTERNAL
120Ω TERMINATION METHOD
BEFORE COMMISSIONING.

DO NOT ENABLE MORE THAN TWO TERMINATION RESISTORS ON THE SAME RS-485 BUS.

D. TB-COM TERMINAL ALLOCATION

TB-COM TERMINAL ALLOCATION				
Terminal	Signal	From	To	Cable
TB-COM:01	RS-485 A	COM-01 A	PLC-01 A	CG-05
TB-COM:02	RS-485 B	COM-01 B	PLC-01 B	CG-05
TB-COM:03	Signal Ground / SG	COM-01 SG	PLC-01 SG	CG-05
PE-BAR	Cable Shield	CG-05 Shield Drain	PE-Bar	CG-05

[GENERAL COMMUNICATION NOTES]

[1] MCU-01 SHALL OPERATE AS MODBUS RTU MASTER.
PLC-01 SHALL OPERATE AS MODBUS RTU SLAVE, STATION ID 1.

[2] PLC-01 AND MCU-01 COMMUNICATION PARAMETERS SHALL BE:
9600 BAUD, 8 DATA BITS, NO PARITY, 1 STOP BIT.

[3] RS-485 A/B POLARITY SHALL BE VERIFIED AGAINST THE
ACTUAL PLC-01 AND COM-01 TERMINAL LABELS BEFORE COMMISSIONING.

[4] RS-485 TERMINATION SHALL BE ENABLED ONLY AT THE TWO
PHYSICAL ENDPOINTS OF THE BUS.

[5] CG-05 SHALL USE SHIELDED TWISTED PAIR CABLE.
CABLE SHIELD SHALL BE BONDED TO PE-BAR AT PANEL END ONLY.

[6] TB-COM:03 SIGNAL GROUND IS A COMMUNICATION REFERENCE.
IT SHALL NOT BE USED AS CABLE SHIELD CONNECTION.

[7] D0020 TEST_RESULT_CODE, D0021 TEST_COMPLETE_FLAG,
D0022 TEST_REQUEST_FLAG, AND D0023 HEARTBEAT COUNTER:
SEE DWG 011 / SHT 2 OF 2 AND DWG 013 / SHT 2 OF 4.

[8] COMMUNICATION WATCHDOG FAULT SHALL BE HANDLED AS
ALM-108. SEE DWG 012.

NO.	DATE	REVISION
A	2026-09-16	INITIAL RELEASE

DWG TITLE

EDS Wafer Map Linked Package
Final Test and Multi-Bin Sorting Cell

MODBUS RS-485 COMMUNICATION CIRCUIT

DRAWN BY: J. H. KO
DATE: 2026-09-16
DWG NO: 014
SHEET NO: 1 OF 1